CLASS 9 · CBSE · GANITA MANJARI PART 1 · FORMULA BANK

Chapter 2: Introduction to Linear Polynomials

Every formula and result of the chapter on one sheet

1	Linear polynomial	$ax + b$, with $a \neq 0$
2	Degree	The highest power of the variable
3	Names by degree	0 constant, 1 linear, 2 quadratic, 3 cubic
4	Linear equation	A linear polynomial set equal to a number, such as $2x + 10 = 64$
5	Linear pattern, nth term	(common difference) $\times\, n +$ constant
6	Linear growth	Starting value $+$ (fixed increase) $\times$ (number of intervals)
7	Linear decay	Starting value $-$ (fixed decrease) $\times$ (number of intervals)
8	Linear relationship	$y = ax + b$
9	Slope	a in $y = ax + b$
10	y-intercept	b in $y = ax + b$; the line cuts the y-axis at $(0, b)$
11	Line through the origin	$y = ax$
12	Parallel lines	Same slope a, different y-intercepts b
13	Rising and falling lines	Slope positive: rises to the right. Slope negative: falls to the right.